

MOHAMED ELSAID

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PROFESSIONAL SUMMARY

Pharmaceutical scientist with an M.S. in Bioinformatics (Johns Hopkins, 2026) and 3+ years of industry experience at Pfizer and Catalent. Builds NGS pipelines and machine learning workflows in Python, R, SQL, and Bash, applying GMP/GLP-grade data integrity and documentation standards to computational biology. Seeking an entry-level Bioinformatics Analyst, Computational Biology, or Biotech Data Analyst role where scientific rigor and reproducibility are valued.

TECHNICAL SKILLS

- **Programming:** Python, R, SQL, Bash, Linux/Unix, Git
- **Bioinformatics:** RNA-seq, NGS Pipelines, Variant Calling, STAR, GATK4, SAMtools, FastQC, DESeq2, GO/KEGG, Snakemake
- **Data Science:** scikit-learn, Random Forest, SVM, Logistic Regression, Statistical Modeling, Data Visualization
- **Laboratory:** GMP, GLP, LIMS, ALCOA+, Data Integrity, 21 CFR Part 11, SOP Development

PROFESSIONAL EXPERIENCE

Associate Scientist, QC Microbiology | Catalent Pharma Solutions — Harmans, MD

Apr 2025 – Oct 2025

- Executed GMP analytical workflows processing 200+ samples per week for sterility and bioburden testing; applied measurement uncertainty quantification to identify out-of-specification trends and support batch disposition.
- Enforced ALCOA+ data integrity standards across the full sample lifecycle via LIMS-based workflows and audit trail review, achieving zero data integrity findings over a 6-month GMP period.
- Authored and revised 10+ SOPs and technical reports compliant with 21 CFR Part 11; led a documentation redesign adopted department-wide that improved audit readiness and reduced review cycles.
- Coordinated cross-functional response among Manufacturing, QA, and R&D to resolve instrumentation failures and deliver analytical results under time-critical batch release constraints.

Associate Scientist | Pfizer Inc. — Pearl River, NY

Aug 2022 – Feb 2024

- Supported high-throughput immunogenicity testing for SARS-CoV-2 vaccine programs, executing quantitative data collection and assay analysis across 800–1,000 samples per week over 18 months of active vaccine development.
- Performed statistical analysis and variability assessments across 15+ large-scale biomedical datasets using R; produced analytical data packages directly supporting FDA and international regulatory submissions.
- Redesigned sample processing workflows with calibration controls and quality checkpoints, reducing analytical turnaround time by ~15% and eliminating recurring manual bottlenecks.
- Authored 20+ GMP-grade SOPs, method validations, and analytical reports supporting clinical operations, QA, and laboratory management stakeholders.

BIOINFORMATICS PORTFOLIO

Public, version-controlled bioinformatics and ML portfolio spanning NGS pipeline engineering and applied machine learning — full project map, code, and documentation at mohamedelsaid-bit.github.io/Portfolio-.

- Built and CI-validated a leakage-aware ML classification pipeline (Random Forest, SVM, Logistic Regression) with 5-fold cross-validation, reaching ~98% held-out test accuracy; documented limitations and reproducibility explicitly.
- Designed a Snakemake-orchestrated RNA-seq differential expression workflow (FastQC → Trimmomatic → STAR → featureCounts → DESeq2 → GO/KEGG enrichment); pipeline execution and results in progress.
- Actively developing a GATK4 Best Practices germline variant calling pipeline (BWA-MEM2, SnpEff) and a Python full-stack FASTA/GC-content analysis tool with SQL integration and Linux deployment.

EDUCATION

M.S. in Bioinformatics — Johns Hopkins University, Baltimore, MD

Sep 2024 – May 2026

B.S. in Biochemistry — Manhattan College, Bronx, NY

Sep 2018 – May 2022

Note: Oct 2025 – May 2026 dedicated to full-time M.S. completion at Johns Hopkins University.